



E.1 Structure of the atom

Nuclear and quantum physics · IB Diploma · Concept guide and original exam-style practice

Understand the concept

The nuclear model of the atom, established by Rutherford's alpha-particle scattering experiment, places nearly all the atom's mass and its positive charge in a tiny, dense central nucleus, with electrons occupying the much larger surrounding volume; the large-angle scattering of some alpha particles indicated a small, concentrated positive charge and repulsion via the Coulomb force. A nuclide is specified by its atomic number Z (protons) and mass number A (protons + neutrons), written as superscript A , subscript Z before the element symbol; isotopes share Z but differ in neutron number. Atomic emission and absorption spectra consist of discrete lines because electrons occupy quantised energy levels, and a photon is emitted or absorbed when an electron transitions between levels, with photon energy $E = hf = E_{\text{initial}} - E_{\text{final}}$ matching the energy gap exactly.

Worked example: follow the reasoning

An electron in a hydrogen atom drops from an energy level of -1.51 eV to a level of -3.40 eV. Find the frequency of the emitted photon. ($h = 6.63 \times 10^{-34}$ J·s, $1 \text{ eV} = 1.60 \times 10^{-19}$ J)
 $\Delta E = E_{\text{initial}} - E_{\text{final}} = (-1.51) - (-3.40) = 1.89 \text{ eV} = 1.89 \times 1.60 \times 10^{-19} = 3.024 \times 10^{-19} \text{ J}$.
 $f = \Delta E/h = 3.024 \times 10^{-19} / 6.63 \times 10^{-34} = 4.56 \times 10^{14} \text{ Hz}$.
Answer: $f \approx 4.56 \times 10^{14} \text{ Hz}$.

Quick check · multiple choice

Rutherford's alpha-particle scattering experiment showed that a small fraction of alpha particles were scattered through large angles, some almost back the way they came. What did this indicate about the atom?

- A The atom is a diffuse cloud of positive charge
- B Most of the atom's mass and positive charge is concentrated in a tiny nucleus
- C Electrons are located inside the nucleus
- D Atoms have no internal structure

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A particular atomic emission spectrum shows a spectral line corresponding to a photon of wavelength 486 nm . ($h = 6.63 \times 10^{-34} \text{ J}\cdot\text{s}$, $c = 3.00 \times 10^8 \text{ m}\cdot\text{s}^{-1}$, $1 \text{ eV} = 1.60 \times 10^{-19} \text{ J}$) [7]
- (a) Calculate the energy of this photon in joules.
 - (b) Express this energy in electron volts.
 - (c) An electron transition in the atom producing this photon occurs from an energy level of -0.85 eV . Calculate the energy of the level to which the electron falls.

Final answer: _____



Answers and explanation

Question 1 [7]

(a) $E = hc/\lambda = (6.63 \times 10^{-34})(3.00 \times 10^8)/(486 \times 10^{-9}) = (1.989 \times 10^{-25})/(4.86 \times 10^{-7}) = 4.09 \times 10^{-19} \text{ J}$.

(b) $E(\text{eV}) = 4.09 \times 10^{-19} / 1.60 \times 10^{-19} = 2.56 \text{ eV}$.

(c) $E_{\text{initial}} - E_{\text{final}} = 2.56 \text{ eV} \rightarrow E_{\text{final}} = E_{\text{initial}} - 2.56 = -0.85 - 2.56 = -3.41 \text{ eV}$.

Quick check answer

B. Large-angle scattering requires a strong, concentrated repulsive force from a small, massive, positively charged nucleus rather than a spread-out charge distribution.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Physics HL/SL subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.